Mosab Ibrahim

Software & Systems Engineer · Infra · SRE

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Software & Systems Engineer grounded in Infra, Reliability Engineering, and Distributed Systems, with deep experience managing database infrastructure across some of the largest fleets on the planet, at ***Meta***, ***Twitter***, and ***Databricks***, spanning millions of customer databases and billions of users. Experience also spans online recruitment platforms, fintech, adtech, and other industries, from early-stage startups to hyperscale, backed by a track record of mentoring engineers, improving processes, and building consensus and alignment within teams and across orgs.

Work Experience

Databricks / Neon.com

Remote, United Kingdom

Senior Production Engineer, Lakebase

Apr 2025 - Sep 2026

- Reliability engineer for Lakebase, Databricks' Serverless Postgres for apps and agents (built on Neon, which Databricks acquired), operating and expanding a multi-region, multi-cloud fleet hosting millions of customer databases.
- Designed and implemented a fleet-wide pod-level network-isolation project across AWS, Azure and GCP. This was a security control required to unblock the platform's public availability, and included deep dives into and tuning of different CNI implementations (cutting CPU by 97% and memory by 99%, and reducing policy-reconcile latency from tens of seconds to tens of milliseconds).
- Integrated Neon and Lakebase with existing Databricks platform services and onboarded Lakebase teams onto them, building a healthy feedback loop with the cross-org teams Lakebase depends on through contributed proposals, fixes and tooling.
- Led shaping the incident response process and best practices, and participated in critical team and org level on-call rotations leading investigations and authoring postmortems for high-blast-radius, multi-region outages, driving root-cause and incident review meetings, including commanding multiple concurrent SEV0 incidents in a single on-call shift.
- Key contributor to the observability and alerting strategies, and best practices for on-call engineers on setting up monitoring dashboards & alerts. Also drove multiple self-initiated cross-team and cross-org efforts to improve incident/alert taxonomy and SLIs/SLOs and define them as code.
- Reduced on-call toil through automation and tooling; implemented a unified on-call CLI, automated cloud health-event ingestion with auto-remediation, and Terraform drift detection.
- Addressed a range of platform security findings, hardening its service security posture.

Technical Skills: Go, Python, Rust, PostgreSQL, Kubernetes, Terraform, AWS, Azure, GCP, Prometheus, Grafana, Distributed Systems

Career Break

United Kingdom

Family & Personal Care

Sep 2024 - Apr 2025

- Stepped away for family and personal reasons, and returned recharged and ready to build.

Meta / Facebook

London, United Kingdom

Production Engineer

Mar 2022 - Aug 2024

- Member of the Backup and Restore team, building services and tooling to recover data from backups for MySQL and ZippyDB (Meta's main key/value store) reliably, efficiently and correctly, serving as the defence line against data loss and corruption across Facebook, Messenger, Instagram and WhatsApp (3.7 billion users). MySQL at Meta is one of the largest installations in the world.
- Focused on the recovery area, leading projects that unblocked large-scale data restoration in disaster scenarios and reduced RTO (Recovery Time Objective) for the datastores we supported.
- Identified performance bottlenecks in fleet-wide backup and restore agents and implemented fixes that reduced their memory and compute footprint across hundreds of thousands of servers.
- Participated in on-call rotation and led investigations into production incidents (including high-visibility SEVs), driving them to mitigation, root cause and tracked follow-ups, and helping other teams recover using their backups. Introduced investigation best practices adopted across teams, and improved monitoring dashboards and alerting to surface issues earlier.
- Designed and implemented integrations with the platform services our systems depend on (e.g. the different services/tiers of storage systems where backups were kept), and led and participated in decommissioning projects affecting teams across Meta, ensuring they were carried out responsibly and reliably.
- Mentored and onboarded engineers into the team's stack, practices and culture, and influenced roadmap and prioritisation by building consensus and deriving insights from production issues and toil.

Technical Skills: Python, Rust, Go, Linux, MySQL, ZippyDB, Distributed Systems, Database Infra, Thrift, Mercurial

Twitter

London, United Kingdom

Senior Site Reliability Engineer

May 2020 - Mar 2022

- Member of Twitter's Database SRE team providing the entire company with database infra, mainly MySQL clusters (both dedicated and multi-tenant), in addition to PostgreSQL, Vertica, and Druid clusters.
- Designed and built a service that automated the team's provisioning toil, cutting time-to-provision by 98% and saving over 100k person-hours (roughly 11.5 person-years) over the following years.
- Monitored and improved availability, scalability, performance and reliability of services owned by our team.
- Participated in the team's on-call rotation, ensuring production issues and incidents are uncovered and addressed properly.
- Led projects to reduce on-call noise, and authored runbooks and documentation improving the team's incident response posture.
- Mentored and onboarded new team members to the team, the tech stack, and helped them become effective and successful members on the team.

Technical Skills: Python, Flask, SQLAlchemy, Linux, MySQL, PostgreSQL, Vertica, Druid, Apache Mesos, Apache Aurora, Git

GoCardless

London, United Kingdom

Site Reliability Engineer

Mar 2019 - Mar 2020

- Protected, provided for, and progressed the software and systems that make up the GoCardless product, supporting product teams in the deployment of high-quality services.

Technical Skills: Go, PostgreSQL, GCP, Kubernetes, Terraform, Helm, Elasticsearch, Kibana, Linux, Docker, Git, Prometheus, Grafana, Thanos

WeQ Global Tech GmbH

Berlin, Germany

Software Engineer

Nov 2017 - Mar 2019

- Developed, deployed and maintained customer-facing services handling billions of requests per day.
- Created and maintained infrastructure as code for both new and legacy systems/microservices.
- Implemented autoscaling and self-healing features into new services.

Technical Skills: PHP, Java, Spring Boot, Vert.x, Kafka, Linux, MySQL, MongoDB, AWS, Docker, Docker Swarm, Terraform, Vagrant, Jenkins, Prometheus, Grafana

Levelset (formerly zlabs, acquired by Procore)

Cairo, Egypt

Senior Software & Platform Engineer - Tech Lead

Jan 2017 - Sep 2017

- Led the team in designing and developing an internal platform for data verification and research process.
- Led the migration of the levelset.com platform to a microservices based architecture, and mentored the team throughout the process.
- Designed and built the API that enabled Levelset to onboard enterprise customers by bulk importing their projects into the platform.
- Worked with enterprise customers to integrate their systems with our API and provided them with the support they needed to successfully onboard to the platform.
- Designed and built the PDF generation engine that generated legal/financial letters for all our customers to secure their lien rights. This is the core feature of the platform.
- Designed and led building various growth products/features that acquired new customer segments.

Senior Software Engineer, DevOps

Mar 2013 - Jan 2017

- Overhauled, stabilised and scaled the MySQL backing infrastructure.
- Designed the new cloud infrastructure (AWS), built automation tools and established system monitoring, performance measurements, and fully automated deployments.
- Migrated the platform from single point of failure virtual private servers (Slicehost) to the cloud (AWS).
- Worked closely with engineers and business folks to identify and solve operational and customer issues with our services, tools and apps.
- Optimised time to release by orders of magnitudes (from 2 days to an average of 3-5 minutes), and improved uptime (99.99% from several outages a week).
- Maintained and operated Git source repositories, defining branching strategies and tutoring the engineering teams about it.

Technical Skills: PHP, Symfony, Linux, MySQL, Neo4j, AWS, RabbitMQ, SOAP API, REST API, Jenkins, Apache Web Server, Git, Nagios, New Relic

Handmade Studios

Cairo, Egypt

Senior Software Engineer

Jun 2012 - Feb 2013

- Designed, developed and maintained an editorial website for entrepreneurs and startups.

Technical Skills: PHP, Symfony, Linux, MySQL, MongoDB, AWS, Apache Web Server, Git

Inmobly

Cairo, Egypt

Senior Software Engineer

Nov 2011 - May 2012

- Designed, developed and deployed API backends for two mobile apps that recorded and streamed videos.

Technical Skills: Python, Flask, SQLAlchemy, REST, AWS, Subversion (svn)

Bluw i LTD

Cairo, Egypt

Technical Consultant

Jun 2011 - Sep 2011

- Implemented a social network for entrepreneurs to raise funds from angel investors.

Technical Skills: PHP, Drupal, AWS, Apache Web Server, Git

Wuzzuf.net

Cairo, Egypt

Software Engineer, Tech Lead

Apr 2010 - Apr 2011

- Designed the architecture and laid out the technical vision and guidelines for Wuzzuf.net, the leading platform in Egypt and the Middle East.
- Worked with, guided, and mentored the team in executing the technical vision, and delivering Wuzzuf.net from scratch.
- Reported directly to the CTO and CEO.
- Designed, created, monitored and maintained the infrastructure of Wuzzuf.net.

Technical Skills: PHP, MySQL, AWS, Apache Web Server, Mercurial, CakePHP, Doctrine ORM, BitBucket

Arabia Inform

Cairo, Egypt

Senior Software Engineer

Nov 2009 - Apr 2010

- Designed and built an in-house media archival and indexing solution with media encoding and advanced search capabilities and features.
- Implemented and launched ads management infra and services for the company's websites network.
- Mentored engineers from other teams throughout critical zero downtime infra upgrades and migration projects.

Technical Skills: PHP, Drupal, MySQL, Apache Web Server, OpenX, AWS, Git

Citex Software

Cairo, Egypt

Software Engineer

Oct 2008 - Nov 2009

- Developed and shipped a social platform that connected entrepreneurs with angel investors to raise funding.

Technical Skills: PHP, Drupal, MySQL, AWS, Git, Apache Web Server, Nagios

Media International

Cairo, Egypt

PHP Developer

Jan 2007 - Sep 2009

- Designed, developed, and deployed various PHP projects for customers and maintained them in production ensuring their availability.
- Led various data migration projects for clients, ensuring safe, correct, and efficient data migration from legacy systems to the new solutions we implemented with minimal to no downtime.
- Created and maintained the "Tech Radar" for the org.
- Designed and implemented load balancing for web services.
- Designed and implemented MySQL architecture, including replication.
- Implemented a service to centralise file and object management solution for our services.
- Designed and implemented the monitoring strategies and infra for our services, and was heavily involved in resolving production issues, and maintaining our services availability and uptime.

Technical Skills: Linux, MySQL, PHP, Drupal, Joomla, Moodle, Apache Web Server, Nagios

20at.com

Cairo, Egypt

PHP Developer

Jun 2004 - Dec 2006

- Designed and implemented a content management system (CMS) from scratch to meet the demands and requrements of the editorial and business teams.
- Designed and executed data migration projects from the legacy systems to the newly developed services with minimal downtime.
- Maintained the online services and website and defined the strategies and tooling for bug fixing and security patches.
- Developed and shipped features (including whole new sub-portals) to meet the business and editorial needs for major and current events (e.g. new sub-portals for reporting on Football World Cup 2006, presidential elections in Egypt, and some of the major conflicts in the area).

Technical Skills: PHP, Linux, MySQL, Apache Web Server, S/FTP

Open Source Projects

Toggl Target

github.com/mos3abof/toggl_target

2013

- A CLI tool that helps plan weekly working hours on top of Toggl time tracking.
- Featured in Toggl's official documentation as a reference API client implementation in Python.

Technical Skills: Python, Requests

Grid Status Now

github.com/mos3abof/gridstatusnow

2014

- A service that monitored Egypt's power grid load status, and visualised the data in a useful way. The project exposed the data via a RESTful API.

Technical Skills: PHP, Symfony

Power Close

github.com/mos3abof/power-close

2014

- A browser add-on to close tabs based on a domain name or a keyword that is part of the URL.

Technical Skills: JavaScript, Firefox, Chrome

More Projects

mosab.co.uk/projects/

Education

University of Cairo

Cairo, Egypt

Bachelor's in Computer Engineering [Dropped Out]

Fall 2003 - Summer 2007

Attended the university of Cairo for a few years where I got the chance to explore several disciplines before being forced to drop out due to an unexpected major life event and family commitments. However, I do take pride in being an eager lifelong learner. Here is a list of those disciplines in chronological order:

- Architecture, *1 semester*
- Civil Engineering, *2 semesters*
- Computer Engineering, *4 semesters*
- Commerce, *2 semesters*

Languages

English Professional proficiency

Arabic Native proficiency